JUSTIN TEH

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OBJECTIVE

To grow as a technically grounded and service-oriented professional by contributing to public-sector innovation at the intersection of technology, analytics, and systems thinking. I hope to apply my skills in computer science, AI, data analytics, visualisation, and user-centred problem solving to support evidence-based decision-making, operational excellence, and initiatives that create meaningful impact for Singapore.

EDUCATION

September 2024 – Present
[Graduating in May 2028]

Bachelor of Engineering (Computer Science and Design)
Singapore University of Technology and Design (SUTD)

- Obtained a cGPA of 4.85 out of 5.0 in Freshmore Year.
- Read modules with "Excellent" in Design Thinking and Innovation, Science for a Sustainable World, Data Driven World, Design Energy Systems.

April 2019 – April 2022

Diploma in Biomedical Engineering
Temasek Polytechnic (TP)

- Obtained a cGPA of 3.91 out of 4.0 in course of study.
- Read modules with "Excellence" in Electronic Prototyping, Microcontroller Technology, and Audiometry and Hearing Devices.

SKILLS

TECHNICAL SKILLS

Software Development & Programming

- Proficient in **C++ (Arduino/ESP32), Python, Java** with working knowledge of object-oriented programming.
- Developed full-stack web applications using **ASP.NET MVC** and IoT-integrated application development, utilising **HTML, CSS, JavaScript** syntax.
- Experience with computer vision and augmented reality using OpenCV for ADL monitoring and data capture.

Data, Algorithms & AI Foundations

- Foundation in data structures, sorting algorithms, recursion, basic UML, and complexity analysis.
- Utilised Seaborn, Matplotlib, KNIME for data analytics and visualisation.
- Implemented introductory **machine learning** methods, including linear regression and logistic regression, with evaluation using metrics such as confusion matrix, accuracy, recall, mean squared error, and correlation.

Tools & Environments

- **IDEs**: Visual Studio Code, PyCharm, Code::Blocks, MPLAB (Microchip PIC), Rhino.
- **Circuit Design**: Exposed to EAGLE for PCB schematic and layout design.
- **Hardware**: Skilled in through-hole soldering techniques.

CAD & Prototyping

- Proficient in Fusion **360 CAD** software for small-scale 3D modelling (25x25x20 cm).
- Hands-on experience with 3D printing for rapid prototyping.

Productivity & Communication Tools

- Advanced user of Microsoft Office Suite (Word, Excel, PowerPoint) for presentations, reports, and administrative tasks in academic and co-curricular settings.

Academic Foundation

- Achieved 'Excellent' in most modules of SUTD's interdisciplinary Freshmore and Biomedical Engineering courses.
- Strong theoretical and applied understanding across science, engineering, and design fundamentals.

GENERIC SKILLS

- **Teamwork & Leadership**: Supported the delivery of CCA induction camps and large-scale arts concerts, and helped lead a **100-member media and productions club** during COVID-19 transition by strengthening video production and livestream initiatives.

- **Project Management**: Coordinated programme design, logistics, publicity, and approvals for SUTD's sophomore commencement, contributing to the execution of a new celebration programme with a **60-member team**.
- **Communication**: Applied clear presentation and coordination skills across academic, leadership, and event settings to engage **diverse stakeholders effectively.**
- **Problem Solving**: Used self-directed learning to strengthen technical solutions, including **rapid full-stack upskilling** at A*STAR IME and independent improvement of programming approaches.
- **Adaptability**: Managed competing academic, work, and production commitments while contributing to research, prototyping, and innovation projects, including **an IP2SG patent filing**.

WORK (DESIGN/PROJECT)

Apr 2022 – Aug 2022

Singapore University of Technology and Design
University Module Projects

- Utilised Data Structure & Algorithm concepts such as Hashmaps, ordered-statistics tree, dictionaries of objects, to propose a full-suite reward system for Algorithm module.
- Implemented multi-linear regression machine learning pipeline with an R^2 of 0.8516, MSE of 1475 and MAE of 31.7 after 3 iterations of parameter hypertuning.

Sep 2021 – Apr 2022

A*STAR, Institute of Microelectronics
Software Application Intern

- Apply industrial-standard Full Stack Web Development knowledge using Model-View-Controller (MVC) framework from ASP.NET Core 5 and MQTT) Publisher-Subscriber, integrating the IoT devices.

Apr 2021 – Aug 2022

Temasek Polytechnic
Major Project

- Objective Activities of Daily Living (ADL) Data Acquisition Major Project using Python libraries such as OpenCV, Pygame, Tkinter, SQLite3.

AWARDS/ACHIEVEMENTS

2025

Singapore University of Technology and Design

- Jyoti and Aditya Mathur Student Service Excellence Award AY25/26
- Honours List Award AY24/25
- Translating Innovative Intelligent Design to Entrepreneurship (TIIDE) 2025 Scholar

2022

Temasek Polytechnic

- Smitech Asia Prize
- Temasek CCA Silver Award
- SG Provisional Patent filed for Major Project (IP Number: 10202105829Q)

2021

- CCA Dedication & Outstanding CCA Contributions Award under TP DMPC
- Outstanding CCA Contributions Award under TP DMPC
- Partook in ARTSIC Competition 2021 (Technology Category)

2020

- Held office as Vice President in TP DMPC
- A*STAR Science Award (Polytechnic)

REFERENCES

Ms Siang Chuei Koo
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